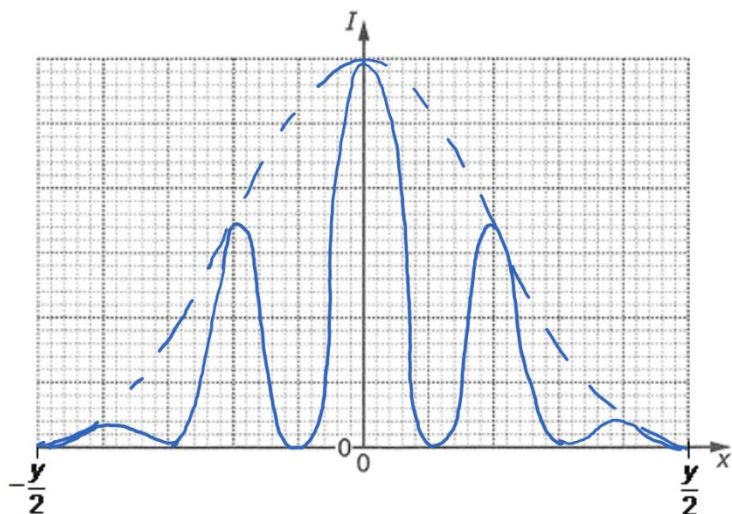


6ai	Diagram must show: (1) Wavefront curving into geometric shadow with some spreading (2) Wavelength unchanged (spacing between wavefronts about equal) and 4 wavefronts drawn	M1 A1
aii	<u>Angle</u> of diffraction increases or More spreading of the wave.	B1
bi	$\sin \theta = \frac{\lambda}{b}$ $\sin \theta = (4.5 \times 10^{-7}) / (0.50 \times 10^{-3})$ Let width be y. $\tan \theta = 0.5 y / 2.5$ $y = 4.500 \text{ mm} = 4.5 \text{ mm}$ Accept small angle approximations, $\sin \theta \approx \tan \theta$	B1 M1 A0
bii1	<u>Coherent waves (from laser)</u> diffracted from the slits <u>will overlap and interfere.</u> When waves meet (in phase) with phase difference of $2n\pi$ rad OR with path difference of $n\lambda$, where $n = 0, \pm 1, \pm 2, \dots$, <u>constructive interference</u> occurs and (maximum intensity is observed). And at the points where the waves (meet anti-phase) with phase difference of $(2n - 1)\pi$ rad OR with path difference of $(n + \frac{1}{2})\lambda$, <u>destructive interference</u> occurs and (minimum intensity) is observed.	B1 B1 B1
bii2	fringe separation, $x = \lambda D / a$ $= (4.5 \times 10^{-7} \times 2.5) / 1.25 \times 10^{-3}$ $= 9.00 \times 10^{-4} = 0.90 \text{ mm}$ Max number of bright fringes on either side of central bright fringe $= 2.25 / 0.9 = 2.5 = 2$ Hence number of bright fringes $= 2 + 2 + 1$ $= 5$	B1 M1 A0

bii3



Intensity of maxima varies according to single diffraction envelope.
 Correct number of fringes
 Reasonable smooth curve with equal fringe separation

B1
 B1
 B1

Total: 13

Section B

--	--	--